VIVEKANANDHA SAINIK COACHING CENTER

ALL INDIA SAINIK SCHOOL ENTRANCE EXAMINATION (AISSEE) 2026
CLASS IX MOCK PAPER 05

Time Allowed: 3 Hours (180 Mins) | Maximum Marks: 400 | Total Questions: 150

GENERAL INSTRUCTIONS:

1. This paper consists of 150 Multiple Choice Questions (MCQs) divided into 5 Sections:
   - **Section A: Mathematics** (Q. 1 to 50) — 4 Marks each (200 Marks)
   - **Section B: Intelligence** (Q. 51 to 75) — 2 Marks each (50 Marks)
   - **Section C: English** (Q. 76 to 100) — 2 Marks each (50 Marks)
   - **Section D: General Science** (Q. 101 to 125) — 2 Marks each (50 Marks)
   - **Section E: Social Science** (Q. 126 to 150) — 2 Marks each (50 Marks)
2. All questions are compulsory. There is NO negative marking for wrong answers.
3. Mark your responses clearly on the OMR sheet. Select exactly ONE correct option for each question.

PART A: QUESTION PAPER (Q. 1 TO 150)

SECTION A: MATHEMATICS (Q. 1 - 50) [50 Qs x 4 Marks = 200 Marks]

Q.1. The multiplicative inverse of -7/12 * (-3/14) is :

(a) 8 **(b)** -8
(c) 18 **(d)** -18

Q.2. Solve for x: (3x - 2)/4 - (2x + 3)/3 = 2/3 - x.

(a) 1 **(b)** 2
(c) 3 **(d)** 4

Q.3. The ratio of two adjacent angles of a parallelogram is 3 : 2. The measures of all its angles are:

(a) 108°, 72°, 108°, 72° **(b)** 100°, 80°, 100°, 80°
(c) 120°, 60°, 120°, 60° **(d)** 110°, 70°, 110°, 70°

Q.4. Find the least square number which is exactly divisible by each of the numbers 8, 15, and 20.

(a) 1800 **(b)** 2400
(c) 3600 **(d)** 4900

Q.5. What is the smallest number by which 68600 must be multiplied to get a perfect cube?

(a) 2 **(b)** 5
(c) 7 **(d)** 10

Q.6. Find the value of m if (2/9)^3 × (2/9)^(-6) = (2/9)^(2m - 1).

(a) -1 **(b)** 1
(c) -2 **(d)** 0

Q.7. A shopkeeper marked an item at 2000 and offered successive discounts of 10% and 5%. The final selling price is:

(a) 1700 **(b)** 1710
(c) 1750 **(d)** 1800

Q.8. Find the compound interest on 10,000 for 1.5 years at 10% per annum, compounded half-yearly.

(a) 1,500.00 **(b)** 1,576.25
(c) 1,600.00 **(d)** 1,525.00

Q.9. If a + b = 9 and ab = 20, then the value of a^2 + b^2 is:

(a) 41 **(b)** 51
(c) 61 **(d)** 81

Q.10. Factorise fully: x^4 - 81.

(a) (x - 3)(x + 3)(x^2 + 9) **(b)** (x - 9)(x + 9)
(c) (x - 3)^2(x + 3)^2 **(d)** (x - 3)(x + 3)(x - 9)

Q.11. 12 workers can build a wall in 8 days. How many workers will be required to build the same wall in 6 days?

(a) 14 **(b)** 16
(c) 18 **(d)** 20

Q.12. The parallel sides of a trapezium are 14 cm and 20 cm, and the distance between them is 8 cm. Find its area.

(a) 120 cm² **(b)** 136 cm²
(c) 144 cm² **(d)** 272 cm²

Q.13. A cylindrical container has a radius of 7 cm and a height of 10 cm. Its total surface area is (Use $\pi = 22/7$):

(a) 440 cm² **(b)** 748 cm²
(c) 616 cm² **(d)** 880 cm²

Q.14. In a pie chart, the central angle for a component representing 25% of the total data is:

(a) 45° **(b)** 60°
(c) 90° **(d)** 120°

Q.15. A bag contains 5 red, 4 blue, and 3 green balls. One ball is drawn at random. What is the probability that it is NOT green?

(a) 1/4 **(b)** 3/4
(c) 5/12 **(d)** 1/3

Q.16. A polyhedron has 12 vertices and 20 faces. How many edges does it have?

(a) 28 **(b)** 30
(c) 32 **(d)** 34

Q.17. The sum of three consecutive multiples of 8 is 888. Find the smallest multiple.

(a) 280 **(b)** 288
(c) 296 **(d)** 304

Q.18. The diagonals of a rhombus are 16 cm and 12 cm long. The length of each side of the rhombus is:

(a) 8 cm **(b)** 10 cm
(c) 12 cm **(d)** 14 cm

Q.19. How many natural numbers lie between 25^2 and 26^2?

(a) 49 **(b)** 50
(c) 51 **(d)** 52

Q.20. Subtract 3a(a + b + c) - 2b(a - b + c) from 4c(-a + b + c).

(a) $-3a^2 - 2b^2 + 4c^2 - ab + 6bc - 7ac$ **(b)** $3a^2 + 2b^2 + 4c^2 + ab - 6bc + 7ac$
(c) $-3a^2 + 2b^2 + 4c^2 - ab - 6bc - 7ac$ **(d)** $-3a^2 - 2b^2 - 4c^2 - ab + 6bc + 7ac$

Q.21. Divide $(5x^2 - 25x + 20)$ by $(x - 1)$.

(a) 5x - 4 **(b)** 5x - 20
(c) 5x + 20 **(d)** x - 4

Q.22. A dealer buys a watch for ₹ 1200 and spends ₹ 100 on repairs. He sells it for ₹ 1430. What is his profit percentage?

(a) 8% **(b)** 10%
(c) 12% **(d)** 15%

Q.23. The standard form of 0.0000035 is:

(a) 3.5 × 10^(-5) **(b)** 3.5 × 10^(-6)
(c) 3.5 × 10^(-7) **(d)** 35 × 10^(-6)

Q.24. Name the property used under multiplication: (-4/5) × 1 = 1 × (-4/5) = -4/5.

(a) Closure property **(b)** Commutative property
(c) Role of 1 as multiplicative identity **(d)** Distributive property

Q.25. The dimensions of a cuboid are in the ratio 1 : 2 : 3 and its total surface area is 88 cm². Its volume is:

(a) 24 cm³ **(b)** 48 cm³
(c) 96 cm³ **(d)** 36 cm³

Q.26. Present ages of Anu and Raj are in the ratio 4 : 5. Eight years from now the ratio of their ages will be 5 : 6. Find their present ages.

(a) 16 and 20 years **(b)** 24 and 30 years
(c) 32 and 40 years **(d)** 40 and 50 years

Q.27. Simplify: (25 × t^(-4)) / (5^(-3) × 10 × t^(-8)) [where t ≠ 0].

(a) 125 t^4 / 2 **(b)** 625 t^4 / 2
(c) 625 t^4 / 4 **(d)** 625 t^4

Q.28. Factorise: a^2 - 2ab + b^2 - c^2.

(a) (a - b - c)(a - b + c) **(b)** (a + b - c)(a - b + c)
(c) (a - b - c)(a + b + c) **(d)** (a + b + c)(a - b - c)

Q.29. The point (-4, 0) lies on which axis?

(a) First quadrant **(b)** Second quadrant
(c) x-axis **(d)** y-axis

Q.30. A car takes 2 hours to reach a destination by travelling at a speed of 60 km/h. How long will it take when travelling at 80 km/h?

(a) 1 hour 15 minutes **(b)** 1 hour 30 minutes
(c) 1 hour 45 minutes **(d)** 2 hours

Q.31. If x/3 + 1 = 7/15, then x is equal to:

(a) -8/5 **(b)** 8/5
(c) -4/5 **(d)** -2/5

Q.32. How many sides does a regular polygon have if each of its interior angles is 165°?

(a) 18 **(b)** 20
(c) 24 **(d)** 30

Q.33. If 15% of a group of 400 students prefer cricket, how many students prefer cricket?

(a) 40 **(b)** 50
(c) 60 **(d)** 75

Q.34. Find the value of √(176 + √2401).

(a) 13 **(b)** 14
(c) 15 **(d)** 16

Q.35. Find the cube root of -13824.

(a) -22 **(b)** -24
(c) -26 **(d)** -28

Q.36. The difference between CI and SI on ₹8000 at 10% per annum for 2 years is:

(a) 60 **(b)** 30
(c) 100 **(d)** 120

Q.37. Evaluate (102)^2 using algebraic identities.

(a) 10400 **(b)** 10404
(c) 10408 **(d)** 10204

Q.38. Factorise: x^2 + 7x + 12.

(a) (x + 2)(x + 6) **(b)** (x + 3)(x + 4)
(c) (x + 1)(x + 12) **(d)** (x - 3)(x - 4)

Q.39. The area of a rhombus is 240 cm² and one of the diagonals is 16 cm. Find the other diagonal.

(a) 20 cm **(b)** 25 cm
(c) 30 cm **(d)** 35 cm

Q.40. If the lateral surface area of a cube is 144 cm², find its volume.

(a) 128 cm³ **(b)** 216 cm³
(c) 256 cm³ **(d)** 343 cm³

Q.41. Value of (3^0 + 4^(-1)) × 2^2 is:

(a) 1 **(b)** 4
(c) 5 **(d)** 7

Q.42. If a 4-digit number 31z5 is a multiple of 3, where z is a digit, what is the smallest value of z?

(a) 0 **(b)** 1
(c) 2 **(d)** 3

Q.43. Solve: 0.25(4f - 3) = 0.05(10f - 9).

(a) 0.4 **(b)** 0.5
(c) 0.6 **(d)** 0.8

Q.44. In a kite ABCD, AB = AD and BC = CD. If ∠B = 100°, ∠D = 100°, and ∠A = 70°, find ∠C.

(a) 80° **(b)** 85°
(c) 90° **(d)** 95°

Q.45. If 15 pipes can fill a tank in 40 minutes, how long will 10 pipes take to fill the tank?

(a) 45 minutes **(b)** 50 minutes
(c) 60 minutes **(d)** 75 minutes

Q.46. Factorise: z - 7 + 7xy - xyz.

(a) (1 - xy)(z - 7) **(b)** (1 + xy)(z - 7)
(c) (xy - 1)(z + 7) **(d)** (x - y)(z - 7)

Q.47. An article marked at ₹840 is sold for ₹714. What is the discount percentage?

(a) 12% **(b)** 14%
(c) 15% **(d)** 18%

Q.48. Find three rational numbers between 1/4 and 1/2.

(a) 5/16, 3/8, 7/16 **(b)** 1/3, 1/5, 1/6
(c) 1/8, 2/8, 3/8 **(d)** 5/8, 6/8, 7/8

Q.49. How many square tiles of side 20 cm are required to pave a floor of dimensions 4 m × 3 m?

(a) 250 **(b)** 300
(c) 350 **(d)** 400

Q.50. Find the product: (2a - 3b)(2a + 3b)(4a^2 + 9b^2).

(a) 16a^4 + 81b^4 **(b)** 16a^4 - 81b^4
(c) 8a^4 - 27b^4 **(d)** 16a^2 - 81b^2

SECTION B: INTELLIGENCE (Q. 51 - 75) [25 Qs x 2 Marks = 50 Marks]

Q.51. Thermometer : Temperature :: Odometer : ?

(a) Speed **(b)** Distance
(c) Pressure **(d)** Current

Q.52. Find the missing term in the series: 3, 7, 15, 31, 63, ?

(a) 125 **(b)** 127
(c) 129 **(d)** 131

Q.53. If 'SCIENCE' is coded as 'TDJFODF', how is 'MATHS' coded in that language?

(a) NBUIT **(b)** LZSGR
(c) NBVIS **(d)** NCUJU

Q.54. Choose the odd one out:

(a) 64 **(b)** 125
(c) 216 **(d)** 144

Q.55. Pointing to a photograph, a man said, 'I have no brother or sister, but that man's father is my father's son.' Whose photograph was it?

(a) His nephew's **(b)** His son's
(c) His father's **(d)** His own

Q.56. A person walks 5 km North, turns right and walks 12 km. How far is he from his starting point?

(a) 13 km **(b)** 17 km
(c) 7 km **(d)** 15 km

Q.57. What comes next in the letter series: A, C, F, J, O, ?

(a) S **(b)** T
(c) U **(d)** V

Q.58. In a class of 40 students, Rohan's rank is 15th from the top. What is his rank from the bottom?

(a) 25th **(b)** 26th
(c) 27th **(d)** 24th

Q.59. Complete the analogy: BCD : EFG :: LMN : ?

(a) OPQ **(b)** PQR
(c) NOP **(d)** QRS

Q.60. If '+' means '×', '-' means '÷', '×' means '-' and '÷' means '+', find the value of: 20 + 5 - 4 ÷ 6 × 5.

(a) 24 **(b)** 26
(c) 28 **(d)** 30

Q.61. The numbers on opposite faces of a standard dice always sum up to:

(a) 6 **(b)** 7
(c) 8 **(d)** 9

Q.62. Choose the correct mirror image of the word 'SAINIK' when mirror is placed on its right.

(a) KINIAK **(b)** KINIAK
(c) SINIAK **(d)** KINIAS

Q.63. Find the missing number in the matrix: Row 1: [4, 5, 20], Row 2: [6, 7, 42], Row 3: [8, 9, ?].

(a) 64 **(b)** 72
(c) 81 **(d)** 90

Q.64. Five friends A, B, C, D, E are sitting in a row facing North. C is sitting in the middle. A is to the immediate right of C. B is at one of the extreme ends and D is next to B. Who is at the left extreme end?

(a) A **(b)** B
(c) D **(d)** E

Q.65. Which Venn diagram best represents the relationship between: Animals, Dogs, Cats?

(a) Two separate circles inside a large circle **(b)** Three intersecting circles
(c) Three concentric circles **(d)** Three completely separate circles

Q.66. Which of the following word pairs is different from the rest?

(a) Cow : Calf **(b)** Dog : Puppy
(c) Cat : Kitten **(d)** Horse : Cub

Q.67. 7 : 50 :: 9 : ?

(a) 80 **(b)** 81
(c) 82 **(d)** 83

Q.68. If 'RED' is coded as 27, how is 'BLUE' coded?

(a) 36 **(b)** 38
(c) 40 **(d)** 42

Q.69. What is the angle between the hour hand and minute hand of a clock at 3:30?

(a) 70° **(b)** 75°
(c) 80° **(d)** 90°

Q.70. How many triangles are there in a square divided by both diagonals?

(a) 6 **(b)** 8
(c) 10 **(d)** 12

Q.71. Which word CANNOT be formed using the letters of the word 'CONSTRUCTION'?

(a) NOTION **(b)** CAUTION
(c) COIN **(d)** SUIT

Q.72. Complete the series: Z2Z, Y4W, X8T, W16Q, ?

(a) V32N **(b)** V32M
(c) U32N **(d)** V64N

Q.73. Arrange in a meaningful logical order: 1. Seed, 2. Plant, 3. Fruit, 4. Tree, 5. Flower.

(a) 1, 2, 4, 5, 3 **(b)** 1, 2, 5, 4, 3
(c) 2, 1, 4, 5, 3 **(d)** 1, 4, 2, 5, 3

Q.74. Statements: All mangoes are golden. All golden things are cheap. Conclusions: I. All mangoes are cheap. II. Golden things are mangoes.

(a) Only conclusion I follows **(b)** Only conclusion II follows
(c) Both follow **(d)** Neither follows

Q.75. A square paper is folded twice and a hole is punched in the center. When unfolded, how many holes will appear?

(a) 1 **(b)** 2
(c) 4 **(d)** 8

SECTION C: ENGLISH (Q. 76 - 100) [25 Qs x 2 Marks = 50 Marks]

Q.76. Dr. A.P.J. Abdul Kalam served as which President of India?

(a) 9th **(b)** 10th
(c) 11th **(d)** 12th

Q.77. Where was Dr. Kalam born?

(a) Chennai **(b)** Rameswaram
(c) Madurai **(d)** Trivandrum

Q.78. What main discipline did Dr. Kalam study besides Physics?

(a) Chemical Engineering **(b)** Aerospace Engineering
(c) Civil Engineering **(d)** Computer Science

Q.79. What was the title of the book in which Dr. Kalam outlined his vision for India?

(a) Wings of Fire **(b)** Ignited Minds
(c) India 2020 **(d)** Turning Points

Q.80. In the passage, which word means 'crucial' or 'of central importance'?

(a) Administrator **(b)** Pivotal
(c) Military **(d)** Civilian

Q.81. Fill in the blank with appropriate tense: By next month, we _____ living in this house for ten years.

(a) will be **(b)** will have been
(c) have been **(d)** were

Q.82. Change to Passive Voice: 'The teacher evaluated the answer sheets.'

(a) The answer sheets are evaluated by the teacher. **(b)** The answer sheets were evaluated by the teacher.
(c) The answer sheets had evaluated by the teacher. **(d)** The answer sheets have been evaluated by the teacher.

Q.83. Change to Indirect Speech: He said to me, 'Where are you going?'

(a) He asked me where I was going. **(b)** He told me where I am going.
(c) He asked me where was I going. **(d)** He asked me that where I was going.

Q.84. Fill in the blanks with suitable articles: She is _____ European lady married to _____ Indian doctor.

(a) an, an **(b)** a, an
(c) a, a **(d)** an, a

Q.85. Fill in the blank: He has been suffering from fever _____ last Monday.

(a) for **(b)** since
(c) from **(d)** in

Q.86. Fill in the blank: Walk carefully _____ you should stumble and fall.

(a) unless **(b)** lest
(c) until **(d)** otherwise

Q.87. Choose the appropriate modal: You _____ respect your elders and teachers.

(a) ought to **(b)** might
(c) can **(d)** could

Q.88. Fill in the blank: Neither the teacher nor the students _____ present in the hall.

(a) was **(b)** were
(c) is **(d)** has been

Q.89. Choose the correct question tag: She sings very well, _____?

(a) doesn't she **(b)** isn't she
(c) don't she **(d)** won't she

Q.90. Choose the word SIMILAR in meaning to 'ABUNDANT':

(a) Scarce **(b)** Plentiful
(c) Rare **(d)** Meagre

Q.91. Choose the word OPPOSITE in meaning to 'OPTIMISTIC':

(a) Hopeful **(b)** Pessimistic
(c) Confident **(d)** Cheerful

Q.92. One-word substitution: A person who studies the scientific history of the earth and its rocks:

(a) Botanist **(b)** Geologist
(c) Astronomer **(d)** Zoologist

Q.93. What does the idiom 'To burn the midnight oil' mean?

(a) To waste fuel **(b)** To work or study late into the night
(c) To cause a fire accident **(d)** To sleep early

Q.94. Choose the correctly spelt word:

(a) Accomodation **(b)** Accommodation
(c) Acommodation **(d)** Accommodashen

Q.95. Spot the error: 'One of my friend (A) / has won (B) / the first prize (C) / in painting (D).'

(a) Part A **(b)** Part B
(c) Part C **(d)** Part D

Q.96. Rearrange P, Q, R, S: P: is essential for, Q: good health, R: Regular exercise, S: maintaining.

(a) RPSQ **(b)** PQRS
(c) SQPR **(d)** QRPS

Q.97. What is the collective noun for a group of lions?

(a) Herd **(b)** Pack
(c) Pride **(d)** Flock

Q.98. Choose the correct superlative degree: 'No other metal is as heavy as lead.'

(a) Lead is heavier than any other metal. **(b)** Lead is the heaviest of all metals.
(c) Lead is a heavy metal. **(d)** Very few metals are as heavy as lead.

Q.99. What is the feminine form of 'Fox'?

(a) Mare **(b)** Vixen
(c) Ewe **(d)** Bitch

Q.100. Fill in the preposition: In accordance _____ the new guidelines, the exam will be held online.

(a) to **(b)** with
(c) of **(d)** for

SECTION D: GENERAL SCIENCE (Q. 101 - 125) [25 Qs x 2 Marks = 50 Marks]

Q.101. Which of the following is a non-contact force?

(a) Friction force **(b)** Muscular force
(c) Electrostatic force **(d)** Tension force

Q.102. Pressure is defined as:

(a) Force × Area **(b)** Force / Area
(c) Area / Force **(d)** Mass × Acceleration

Q.103. Which type of friction is the smallest in magnitude?

(a) Static friction **(b)** Sliding friction
(c) Rolling friction **(d)** Limiting friction

Q.104. The pitch of a sound depends directly on its:

(a) Amplitude **(b)** Frequency
(c) Loudness **(d)** Speed

Q.105. The SI unit of frequency is:

(a) Decibel (dB) **(b)** Hertz (Hz)
(c) Meter/second **(d)** Joule

Q.106. During electroplating of copper onto an iron key, the iron key is connected to the:

(a) Positive terminal (Anode) **(b)** Negative terminal (Cathode)
(c) Either terminal **(d)** Ground

Q.107. The angle of incidence is equal to the angle of reflection:

(a) Always **(b)** Sometimes
(c) Under special conditions **(d)** Never

Q.108. Which part of the human eye controls the amount of light entering it?

(a) Cornea **(b)** Iris
(c) Lens **(d)** Retina

Q.109. Which planet is commonly known as the 'Morning Star' or 'Evening Star'?

(a) Mars **(b)** Venus
(c) Jupiter **(d)** Mercury

Q.110. The magnitude of an earthquake is measured on the:

(a) Barometer **(b)** Richter Scale
(c) Thermometer **(d)** Anemometer

Q.111. Which synthetic fibre is known as 'Artificial Silk'?

(a) Nylon **(b)** Rayon
(c) Polyester **(d)** Acrylic

Q.112. Which non-metal is stored under water to prevent spontaneous combustion in air?

(a) Sodium **(b)** Phosphorus
(c) Sulphur **(d)** Carbon

Q.113. Which metal exists in liquid state at room temperature?

(a) Mercury **(b)** Sodium
(c) Lead **(d)** Zinc

Q.114. The slow chemical process of conversion of dead vegetation into coal is called:

(a) Carbonisation **(b)** Distillation
(c) Evaporation **(d)** Refining

Q.115. Which gas is the most effective fire extinguisher for electrical and oil fires?

(a) Oxygen **(b)** Carbon dioxide
(c) Nitrogen **(d)** Hydrogen

Q.116. Acid rain is primarily caused by atmospheric pollution due to gases:

(a) CO_2 and CO **(b)** SO_2 and NO_2
(c) CFCs and CH_4 **(d)** O_3 and N_2

Q.117. Wheat, gram, and mustard are examples of which crop category in India?

(a) Kharif crops **(b)** Rabi crops
(c) Zaid crops **(d)** Cash crops

Q.118. Rhizobium bacteria present in root nodules of leguminous plants help in:

(a) Photosynthesis **(b)** Nitrogen fixation
(c) Transpiration **(d)** Food preservation

Q.119. The Red Data Book contains a comprehensive record of:

(a) Endemic species **(b)** Extinct species
(c) Endangered species **(d)** Exotic species

Q.120. Which cellular organelle is present in plant cells but absent in animal cells?

(a) Mitochondria **(b)** Cell wall
(c) Nucleus **(d)** Cell membrane

Q.121. Hydra reproduces asexually through the process of:

(a) Binary fission **(b)** Budding
(c) Fragmentation **(d)** Spore formation

Q.122. Which hormone regulates glucose levels in human blood?

(a) Thyroxine **(b)** Adrenaline
(c) Insulin **(d)** Growth hormone

Q.123. The sex chromosome combination present in a human male child is:

(a) XX **(b)** XY
(c) YY **(d)** XO

Q.124. Bakelite and Melamine are classic examples of:

(a) Thermoplastics **(b)** Thermosetting plastics
(c) Natural polymers **(d)** Synthetic rubber

Q.125. The unit of calorific value of a fuel is expressed as:

(a) kJ/g **(b)** kJ/kg
(c) J/m **(d)** Cal/s

SECTION E: SOCIAL SCIENCE (Q. 126 - 150) [25 Qs x 2 Marks = 50 Marks]

Q.126. The historic Battle of Plassey was fought in the year:

(a) 1757 **(b)** 1764
(c) 1857 **(d)** 1761

Q.127. Who led the Revolt of 1857 from Kanpur?

(a) Rani Laxmibai **(b)** Nana Saheb
(c) Kunwar Singh **(d)** Begum Hazrat Mahal

Q.128. Brahmo Samaj was founded in 1828 by:

(a) Swami Dayanand Saraswati **(b)** Raja Ram Mohan Roy
(c) Jyotirao Phule **(d)** Swami Vivekananda

Q.129. The Partition of Bengal was announced in 1905 by Viceroy:

(a) Lord Ripon **(b)** Lord Curzon
(c) Lord Dalhousie **(d)** Lord Mountbatten

Q.130. Gandhiji launched the Civil Disobedience Movement in 1930 with the famous:

(a) Non-Cooperation Movement **(b)** Dandi Salt March
(c) Quit India Movement **(d)** Champaran Satyagraha

Q.131. Resources that are drawn directly from nature and used without much modification are called:

(a) Human-made resources **(b)** Natural resources
(c) Non-renewable resources **(d)** Potential resources

Q.132. Planting rows of trees to check wind movement and protect soil cover is called:

(a) Terrace farming **(b)** Contour ploughing
(c) Shelter belts **(d)** Mulching

Q.133. Which crop is popularly known as the 'Golden Fibre'?

(a) Cotton **(b)** Jute
(c) Silk **(d)** Wool

Q.134. Shifting cultivation is locally known as 'Jhumming' in:

(a) Western Ghats **(b)** North-East India
(c) Thar Desert **(d)** Deccan Plateau

Q.135. Silicon Valley is an important global IT hub located in:

(a) Bangalore **(b)** California, USA
(c) London, UK **(d)** Tokyo, Japan

Q.136. The Constitution of India came into full effect on:

(a) 15th August 1947 **(b)** 26th November 1949
(c) 26th January 1950 **(d)** 2nd October 1950

Q.137. Right to Equality in the Indian Constitution is guaranteed under Articles:

(a) 14 to 18 **(b)** 19 to 22
(c) 23 to 24 **(d)** 25 to 28

Q.138. What is the maximum sanctioned strength of the Lok Sabha under the Constitution?

(a) 245 **(b)** 545
(c) 550 **(d)** 500

Q.139. What does 'PIL' stand for in the Indian legal system?

(a) Public Interest Litigation **(b)** Private Interest Law
(c) Public Information Law **(d)** Police Investigation List

Q.140. Article 17 of the Indian Constitution explicitly abolishes:

(a) Child Labour **(b)** Untouchability
(c) Sati System **(d)** Title Conferment

Q.141. Who introduced the Permanent Settlement system of land revenue in Bengal in 1793?

(a) Warren Hastings **(b)** Lord Cornwallis
(c) Thomas Munro **(d)** Lord Wellesley

Q.142. Wood's Despatch of 1854 was an important official document related to:

(a) Military reforms **(b)** Educational reforms
(c) Police reforms **(d)** Agricultural tax

Q.143. Which of the following is a non-conventional source of energy?

(a) Coal **(b)** Petroleum
(c) Solar Energy **(d)** Natural Gas

Q.144. According to 2011 Census, the average density of population in India was:

(a) 324 persons per sq km **(b)** 382 persons per sq km
(c) 410 persons per sq km **(d)** 450 persons per sq km

Q.145. Under Article 21 of the Constitution, the Right to Clean Water is part of the fundamental:

(a) Right to Freedom **(b)** Right to Life
(c) Right against Exploitation **(d)** Right to Equality

Q.146. Satyashodhak Samaj was founded in 1873 by Jyotirao Phule to promote:

(a) Western education **(b)** Caste equality and rights of lower castes
(c) Widow remarriage only **(d)** Sanskrit revival

Q.147. Which state in India is the leading producer of Iron Ore?

(a) Odisha **(b)** Kerala
(c) Punjab **(d)** Gujarat

Q.148. Who presides over the Joint Sitting of both Houses of Parliament in India?

(a) President of India **(b)** Prime Minister
(c) Speaker of Lok Sabha **(d)** Chairman of Rajya Sabha

Q.149. The Indian National Army (INA) was reorganized in Singapore in 1943 by:

(a) Rash Behari Bose **(b)** Subhas Chandra Bose
(c) Bhagat Singh **(d)** Jawaharlal Nehru

Q.150. The Scheduled Castes and Scheduled Tribes (Prevention of Atrocities) Act was enacted in:

(a) 1980 **(b)** 1989
(c) 1995 **(d)** 2005

PART B: MASTER ANSWER KEY (Q. 1 TO 150)

Q.	Ans	Q.	Ans	Q.	Ans	Q.	Ans	Q.	Ans
1	(a)	31	(a)	61	(b)	91	(b)	121	(b)
2	(b)	32	(c)	62	(d)	92	(b)	122	(c)
3	(a)	33	(c)	63	(b)	93	(b)	123	(b)
4	(c)	34	(c)	64	(b)	94	(b)	124	(b)
5	(b)	35	(b)	65	(a)	95	(a)	125	(b)
6	(a)	36	(b)	66	(d)	96	(a)	126	(a)
7	(b)	37	(b)	67	(c)	97	(c)	127	(b)
8	(b)	38	(b)	68	(c)	98	(b)	128	(b)
9	(a)	39	(c)	69	(b)	99	(b)	129	(b)
10	(a)	40	(b)	70	(b)	100	(b)	130	(b)
11	(b)	41	(c)	71	(b)	101	(c)	131	(b)
12	(b)	42	(a)	72	(a)	102	(b)	132	(c)
13	(b)	43	(c)	73	(a)	103	(c)	133	(b)
14	(c)	44	(c)	74	(a)	104	(b)	134	(b)
15	(b)	45	(c)	75	(c)	105	(b)	135	(b)
16	(b)	46	(a)	76	(c)	106	(b)	136	(c)
17	(b)	47	(c)	77	(b)	107	(a)	137	(a)
18	(b)	48	(a)	78	(b)	108	(b)	138	(c)
19	(b)	49	(b)	79	(c)	109	(b)	139	(a)
20	(a)	50	(b)	80	(b)	110	(b)	140	(b)
21	(b)	51	(b)	81	(b)	111	(b)	141	(b)
22	(b)	52	(b)	82	(b)	112	(b)	142	(b)
23	(b)	53	(a)	83	(a)	113	(a)	143	(c)
24	(c)	54	(d)	84	(b)	114	(a)	144	(b)
25	(b)	55	(b)	85	(b)	115	(b)	145	(b)
26	(c)	56	(a)	86	(b)	116	(b)	146	(b)
27	(b)	57	(c)	87	(a)	117	(b)	147	(a)
28	(a)	58	(b)	88	(b)	118	(b)	148	(c)
29	(c)	59	(a)	89	(a)	119	(c)	149	(b)
30	(b)	60	(b)	90	(b)	120	(b)	150	(b)

PART C: DETAILED STEP-BY-STEP SOLUTIONS (Q. 1 TO 150)

SECTION A: MATHEMATICS (Q. 1 - 50) [50 Qs x 4 Marks = 200 Marks]

Question 1: Correct Option (a)

First, simplify -&frac7;12 × (-&frac3;14) = &frac21;168 = &frac1;8. The multiplicative inverse (reciprocal) of &frac1;8 is 8.

Question 2: Correct Option (b)

Multiply the entire equation by the LCM 12: 3(3x - 2) - 4(2x + 3) = 8 - 12x ⇒ 9x - 6 - 8x - 12 = 8 - 12x ⇒ x - 18 = 8 - 12x ⇒ 13x = 26 ⇒ x = 2.

Question 3: Correct Option (a)

Adjacent angles of a parallelogram are supplementary. 3x + 2x = 180° ⇒ 5x = 180° ⇒ x = 36°. The angles are 3(36°) = 108° and 2(36°) = 72°. Opposite angles are equal, so the four angles are 108°, 72°, 108°, 72°.

Question 4: Correct Option (c)

LCM of 8, 15, 20 = 120 = 2 × 2 × 2 × 3 × 5. To make prime factors pair up, multiply by 2 × 3 × 5 = 30. Required square number = 120 × 30 = 3600.

Question 5: Correct Option (b)

Prime factorisation of 68600 = 2^3 × 5^2 × 7^3. To form complete triplets of prime factors, one more factor of 5 is required (5^2 × 5 = 5^3). Hence, multiply by 5.

Question 6: Correct Option (a)

Using laws of exponents a^m × a^n = a^(m+n): 3 + (-6) = 2m - 1 ⇒ -3 = 2m - 1 ⇒ 2m = -2 ⇒ m = -1.

Question 7: Correct Option (b)

SP = Marked Price × (1 - d1/100) × (1 - d2/100) = 2000 × 0.90 × 0.95 = 1800 × 0.95 = ■1710.

Question 8: Correct Option (b)

P = ■10,000, rate r = 5% per half-year, time n = 3 half-years. Amount A = 10000(1 + 5/100)^3 = 10000(1.05)^3 = ■11,576.25. CI = A - P = ■1,576.25.

Question 9: Correct Option (a)

Using the identity (a + b)^2 = a^2 + b^2 + 2ab: 9^2 = a^2 + b^2 + 2(20) ⇒ 81 = a^2 + b^2 + 40 ⇒ a^2 + b^2 = 41.

Question 10: Correct Option (a)

x^4 - 81 = (x^2)^2 - 9^2 = (x^2 - 9)(x^2 + 9) = (x - 3)(x + 3)(x^2 + 9).

Question 11: Correct Option (b)

Work is in inverse proportion: x1 × y1 = x2 × y2 ⇒ 12 × 8 = x2 × 6 ⇒ x2 = 96 / 6 = 16 workers.

Question 12: Correct Option (b)

Area of trapezium = &frac1;2 × (sum of parallel sides) × height = &frac1;2 × (14 + 20) × 8 = 34 × 4 = 136 cm².

Question 13: Correct Option (b)

Total Surface Area = 2πr(r + h) = 2 × (22/7) × 7 × (7 + 10) = 44 × 17 = 748 cm².

Question 14: Correct Option (c)

Central angle = (Percentage / 100) × 360° = (25 / 100) × 360° = 90°.

Question 15: Correct Option (b)

Total balls = 5 + 4 + 3 = 12. Non-green balls = 5 + 4 = 9. P(Not green) = 9/12 = 3/4.

Question 16: Correct Option (b)

By Euler's formula for polyhedra, F + V - E = 2 ⇒ 20 + 12 - E = 2 ⇒ 32 - E = 2 ⇒ E = 30 edges.

Question 17: Correct Option (b)

Let the multiples be 8x, 8x + 8, and 8x + 16. Sum = 24x + 24 = 888 ⇒ 24x = 864 ⇒ x = 36. Smallest = 8 × 36 = 288.

Question 18: Correct Option (b)

Diagonals of a rhombus bisect each other at 90°. Half diagonals are 8 cm and 6 cm. Side = √(8^2 + 6^2) = √(64 + 36) = √100 = 10 cm.

Question 19: Correct Option (b)

Between n^2 and (n+1)^2, there are 2n non-square natural numbers. For n = 25, 2n = 2(25) = 50 numbers.

Question 20: Correct Option (a)

Exp 1 = $3a^2 + 3ab + 3ac - 2ab + 2b^2 - 2bc = 3a^2 + 2b^2 + ab + 3ac - 2bc$. Exp 2 = $-4ac + 4bc + 4c^2$. Exp 2 - Exp 1 = $-3a^2 - 2b^2 + 4c^2 - ab + 6bc - 7ac$.

Question 21: Correct Option (b)

Factorise numerator: $5(x^2 - 5x + 4) = 5(x - 1)(x - 4)$. Dividing by (x - 1) leaves 5(x - 4) = 5x - 20.

Question 22: Correct Option (b)

Total Cost Price = 1200 + 100 = ■1300. Selling Price = ■1430. Profit = 1430 - 1300 = ■130. Profit % = (130 / 1300) × 100 = 10%.

Question 23: Correct Option (b)

Move the decimal point 6 places to the right to obtain 3.5. Hence, 0.0000035 = 3.5 × 10^(-6).

Question 24: Correct Option (c)

Multiplying any rational number by 1 yields the same number, demonstrating that 1 is the multiplicative identity.

Question 25: Correct Option (b)

Let dimensions be x, 2x, 3x. TSA = $2(2x^2 + 6x^2 + 3x^2) = 22x^2 = 88 \Rightarrow x^2 = 4 \Rightarrow x = 2$. Dimensions are 2, 4, 6 cm. Volume = 2 × 4 × 6 = 48 cm^3.

Question 26: Correct Option (c)

Let present ages be 4x and 5x. (4x + 8)/(5x + 8) = 5/6 ⇒ 24x + 48 = 25x + 40 ⇒ x = 8. Ages are 32 and 40 years.

Question 27: Correct Option (b)

Numerator = 5^2 × t^(-4). Denominator = 5^(-3) × 2 × 5 × t^(-8) = 2 × 5^(-2) × t^(-8). Expression = (5^(2 - (-2)) / 2) × t^(-4 - (-8)) = (5^4 / 2) × t^4 = 625 t^4 / 2.

Question 28: Correct Option (a)

a^2 - 2ab + b^2 - c^2 = (a - b)^2 - c^2 = (a - b - c)(a - b + c).

Question 29: Correct Option (c)

Any point with a y-coordinate equal to 0 lies on the x-axis.

Question 30: Correct Option (b)

Distance = Speed × Time = 60 × 2 = 120 km. New Time = 120 / 80 = 1.5 hours = 1 hour 30 minutes.

Question 31: Correct Option (a)

x/3 = 7/15 - 1 = -8/15 ⇒ x = 3 × (-8/15) = -8/5.

Question 32: Correct Option (c)

Each exterior angle = 180° - 165° = 15°. Number of sides = 360° / 15° = 24.

Question 33: Correct Option (c)

Number of students = (15 / 100) × 400 = 60.

Question 34: Correct Option (c)

√2401 = 49. √(176 + 49) = √225 = 15.

Question 35: Correct Option (b)

13824 = 2^9 × 3^3. Cube root of 13824 = 2^3 × 3 = 24. Hence, √(-13824) = -24.

Question 36: Correct Option (b)

Difference for 2 years = P(r/100)^2 = 8000(10/100)^2 = 8000(1/100) = ■80.

Question 37: Correct Option (b)

(100 + 2)^2 = 100^2 + 2(100)(2) + 2^2 = 10000 + 400 + 4 = 10404.

Question 38: Correct Option (b)

Find numbers whose product is 12 and sum is 7: 3 and 4. So x^2 + 7x + 12 = (x + 3)(x + 4).

Question 39: Correct Option (c)

Area = &frac1;2 × d1 × d2 ⇒ 240 = &frac1;2 × 16 × d2 ⇒ 8 d2 = 240 ⇒ d2 = 30 cm.

Question 40: Correct Option (b)

Lateral surface area = 4a^2 = 144 ⇒ a^2 = 36 ⇒ side a = 6 cm. Volume = a^3 = 6^3 = 216 cm³.

Question 41: Correct Option (c)

(1 + 1/4) × 4 = (5/4) × 4 = 5.

Question 42: Correct Option (a)

Sum of digits = 3 + 1 + z + 5 = 9 + z. For 9 + z to be divisible by 3, the smallest non-negative digit z is 0.

Question 43: Correct Option (c)

f - 0.75 = 0.5f - 0.45 ⇒ f - 0.5f = 0.75 - 0.45 ⇒ 0.5f = 0.30 ⇒ f = 0.6.

Question 44: Correct Option (c)

Sum of interior angles of a quadrilateral = 360°. 70° + 100° + 100° + ∠C = 360° ⇒ 270° + ∠C = 360° ⇒ ∠C = 90°.

Question 45: Correct Option (c)

Inverse proportion: 15 × 40 = 10 × t ⇒ t = 600 / 10 = 60 minutes = 1 hour.

Question 46: Correct Option (a)

(z - 7) - xy(z - 7) = (1 - xy)(z - 7).

Question 47: Correct Option (c)

Discount = 840 - 714 = ■126. Discount % = (126 / 840) × 100 = 15%.

Question 48: Correct Option (a)

Convert to denominator 16: 1/4 = 4/16 and 1/2 = 8/16. Three rational numbers are 5/16, 6/16 = 3/8, and 7/16.

Question 49: Correct Option (b)

Floor area = 400 cm × 300 cm = 120,000 cm². Tile area = 20 × 20 = 400 cm². Number of tiles = 120,000 / 400 = 300.

Question 50: Correct Option (b)

(2a - 3b)(2a + 3b) = 4a^2 - 9b^2. Then (4a^2 - 9b^2)(4a^2 + 9b^2) = (4a^2)^2 - (9b^2)^2 = 16a^4 - 81b^4.

SECTION B: INTELLIGENCE (Q. 51 - 75) [25 Qs x 2 Marks = 50 Marks]

Question 51: Correct Option (b)

A thermometer measures temperature; an odometer measures distance travelled.

Question 52: Correct Option (b)

Pattern: Multiply by 2 and add 1. 3×2+1=7, 7×2+1=15, 15×2+1=31, 31×2+1=63, 63×2+1=127.

Question 53: Correct Option (a)

Each letter is shifted by +1 forward in the alphabet: M→N, A→B, T⇒U, H⇒I, S⇒T = NBUIT.

Question 54: Correct Option (d)

64 = 4^3, 125 = 5^3, 216 = 6^3 are perfect cubes. 144 = 12^2 is a square number, not a cube.

Question 55: Correct Option (b)

My father's son = the speaker himself (since he has no siblings). 'That man's father is myself' implies the photograph is of his son.

Question 56: Correct Option (a)

By Pythagoras theorem: Shortest distance = √(5^2 + 12^2) = √(25 + 144) = √169 = 13 km.

Question 57: Correct Option (c)

Positional gaps increase by +2, +3, +4, +5, +6. A(1)+2=C(3), C+3=F(6), F+4=J(10), J+5=O(15), O(15)+6=U(21).

Question 58: Correct Option (b)

Rank from bottom = (Total students - Rank from top) + 1 = (40 - 15) + 1 = 26th.

Question 59: Correct Option (a)

Shift each letter by +3: L→O, M→P, N→Q = OPQ.

Question 60: Correct Option (b)

Replacing operators: 20 × 5 ÷ 4 + 6 - 5 = 100 ÷ 4 + 6 - 5 = 25 + 6 - 5 = 26.

Question 61: Correct Option (b)

In a standard dice, the sum of numbers on opposite faces is always 7 (1+6, 2+5, 3+4).

Question 62: Correct Option d

Letters reverse in order and undergo horizontal lateral inversion.

Question 63: Correct Option (b)

Pattern: First number × Second number = Third number. 8 × 9 = 72.

Question 64: Correct Option (b)

Positions 1 to 5: C is at 3. A is at 4. If B were at 5, D would be at 4 (conflict). Hence B is at 1 (left extreme end), D is at 2, C at 3, A at 4, E at 5.

Question 65: Correct Option (a)

Dogs and Cats are disjoint sets, both belonging entirely to the set of Animals.

Question 66: Correct Option (d)

A young horse is called a foal or colt. A cub is the young one of a lion/bear/tiger.

Question 67: Correct Option (c)

Pattern: n^2 + 1. 7^2 + 1 = 50. Similarly, 9^2 + 1 = 82.

Question 68: Correct Option (c)

Sum of alphabetical positions: RED = 18 + 5 + 4 = 27. BLUE = 2 + 12 + 21 + 5 = 40.

Question 69: Correct Option (b)

Angle = |30H - 5.5M| = |30(3) - 5.5(30)| = |90 - 165| = 75°.

Question 70: Correct Option (b)

4 small individual triangles + 4 large triangles formed by combining two adjacent small ones = 8 triangles.

Question 71: Correct Option (b)

The letter 'A' is not present in the word CONSTRUCTION.

Question 72: Correct Option (a)

First letter: Z, Y, X, W, V (-1). Number: 2, 4, 8, 16, 32 (×2). Last letter: Z, W, T, Q, N (-3). Answer: V32N.

Question 73: Correct Option (a)

Logical biological order: Seed (1) → Plant (2) → Tree (4) → Flower (5) → Fruit (3).

Question 74: Correct Option (a)

All mangoes are subset of golden, which is subset of cheap ⇒ All mangoes are cheap (I follows). All golden things are not necessarily mangoes (II invalid).

Question 75: Correct Option (c)

Unfolding a twice-folded paper with a center hole multiplies the hole by 2^2 = 4 times.

SECTION C: ENGLISH (Q. 76 - 100) [25 Qs x 2 Marks = 50 Marks]

Question 76: Correct Option (c)

As stated in the passage, Dr. Kalam was the 11th President of India.

Question 77: Correct Option (b)

According to the passage, Dr. Kalam was born in Rameswaram, Tamil Nadu.

Question 78: Correct Option (b)

The passage mentions that he studied physics and aerospace engineering.

Question 79: Correct Option (c)

Dr. Kalam advocated his vision to transform India into a developed nation in 'India 2020'.

Question 80: Correct Option (b)

'Pivotal' means playing a central, vital, or crucial role.

Question 81: Correct Option (b)

Future Perfect Continuous tense ('will have been living') is required for an action continuing up to a future point in time.

Question 82: Correct Option (b)

Simple Past passive structure: Object + was/were + V3 + by + Subject.

Question 83: Correct Option (a)

Wh-questions do not take 'that'; subject comes before verb in indirect reporting ('where I was going').

Question 84: Correct Option (b)

'European' begins with a consonant sound /j/ ('a'), while 'Indian' begins with a vowel sound ('an').

Question 85: Correct Option (b)

'Since' is used to denote a specific point of starting time in Perfect Tenses.

Question 86: Correct Option (b)

'Lest' expresses fear of an unwanted result and is followed by 'should'.

Question 87: Correct Option (a)

'Ought to' expresses moral duty, obligation, or correctness.

Question 88: Correct Option (b)

When subjects are joined by 'neither... nor', the verb agrees with the plural subject closer to it ('students').

Question 89: Correct Option (a)

Positive Simple Present statement ('sings') takes a negative auxiliary question tag ('doesn't she').

Question 90: Correct Option (b)

'Abundant' means existing or available in large quantities; 'plentiful' is its synonym.

Question 91: Correct Option (b)

'Optimistic' means hopeful/positive; 'pessimistic' means expecting the worst.

Question 92: Correct Option (b)

A geologist studies the solid earth, rocks, and geological history.

Question 93: Correct Option (b)

'To burn the midnight oil' means to study or work hard late into the night.

Question 94: Correct Option (b)

The correct spelling is 'Accommodation' (double 'c' and double 'm').

Question 95: Correct Option (a)

'One of' is always followed by a plural noun ('One of my friends').

Question 96: Correct Option (a)

Correct arrangement: 'Regular exercise (R) is essential for (P) maintaining (S) good health (Q).'

Question 97: Correct Option (c)

A group of lions is called a 'pride'.

Question 98: Correct Option (b)

Superlative form of 'heavy' is 'the heaviest'.

Question 99: Correct Option (b)

The female fox is called a 'vixen'.

Question 100: Correct Option (b)

The correct idiom is 'in accordance with'.

SECTION D: GENERAL SCIENCE (Q. 101 - 125) [25 Qs x 2 Marks = 50 Marks]

Question 101: Correct Option (c)

Electrostatic, gravitational, and magnetic forces act at a distance without physical contact.

Question 102: Correct Option (b)

Pressure is defined as force per unit area (P = F / A).

Question 103: Correct Option (c)

Rolling friction is significantly smaller than static or sliding friction.

Question 104: Correct Option (b)

Pitch is determined by the frequency of vibration; higher frequency produces a higher pitch.

Question 105: Correct Option (b)

Frequency is measured in Hertz (Hz), representing oscillations per second.

Question 106: Correct Option (b)

The object to be electroplated is always connected to the negative terminal (cathode).

Question 107: Correct Option (a)

The Law of Reflection states that angle of incidence is ALWAYS equal to angle of reflection.

Question 108: Correct Option (b)

The iris regulates the size of the pupil, controlling light entry into the eye.

Question 109: Correct Option (b)

Venus shines brightly in the sky before sunrise or after sunset, called Morning/Evening Star.

Question 110: Correct Option (b)

Earthquake magnitude is recorded quantitatively on the logarithmic Richter Scale.

Question 111: Correct Option (b)

Rayon is derived from wood pulp and resembles silk in texture and appearance.

Question 112: Correct Option (b)

White phosphorus is extremely reactive with atmospheric oxygen and is kept immersed in water.

Question 113: Correct Option (a)

Mercury (Hg) is the only metal that is liquid at room temperature.

Question 114: Correct Option (a)

Under high temperature and pressure underground, dead flora turns into coal via carbonisation.

Question 115: Correct Option (b)

Carbon dioxide (CO_2) cuts off oxygen supply and does not conduct electricity.

Question 116: Correct Option (b)

Sulphur dioxide (SO_2) and Nitrogen dioxide (NO_2) react with water vapour forming sulphuric and nitric acids.

Question 117: Correct Option (b)

Rabi crops are sown in winter (October-November) and harvested in spring (March-April).

Question 118: Correct Option (b)

Rhizobium fixes atmospheric nitrogen into usable nitrogenous compounds for plants.

Question 119: Correct Option (c)

Red Data Book published by IUCN maintains records of endangered flora and fauna.

Question 120: Correct Option (b)

Plant cells possess a rigid outer cell wall made of cellulose surrounding the plasma membrane.

Question 121: Correct Option (b)

Hydra develops a small outgrowth called a bud which detaches to form a new individual.

Question 122: Correct Option (c)

Insulin secreted by the pancreas regulates blood sugar level in humans.

Question 123: Correct Option (b)

Human males possess one X chromosome from mother and one Y chromosome from father (XY).

Question 124: Correct Option (b)

Thermosetting plastics cannot be softened by heating once molded into shape.

Question 125: Correct Option (b)

Calorific value is the amount of heat energy produced by complete combustion of 1 kg fuel (kJ/kg).

SECTION E: SOCIAL SCIENCE (Q. 126 - 150) [25 Qs x 2 Marks = 50 Marks]

Question 126: Correct Option (a)

Battle of Plassey took place on 23 June 1757 between East India Company and Siraj-ud-Daulah.

Question 127: Correct Option (b)

Nana Saheb, the adopted son of Peshwa Baji Rao II, led the uprising in Kanpur.

Question 128: Correct Option (b)

Raja Ram Mohan Roy established Brahmo Sabha (later Brahmo Samaj) in 1828 in Bengal.

Question 129: Correct Option (b)

Lord Curzon issued the order to partition Bengal in July 1905.

Question 130: Correct Option (b)

Dandi March (Salt Satyagraha) from Sabarmati to Dandi marked the beginning of Civil Disobedience.

Question 131: Correct Option (b)

Natural resources are provided by nature (air, water, soil, minerals).

Question 132: Correct Option (c)

Shelter belts are rows of trees planted in coastal and arid regions to prevent soil erosion.

Question 133: Correct Option (b)

Jute is termed Golden Fibre due to its silky golden color and high economic value.

Question 134: Correct Option (b)

Slash-and-burn shifting agriculture in North-Eastern states of India is called Jhumming.

Question 135: Correct Option (b)

Silicon Valley is part of the San Francisco Bay Area in California, USA.

Question 136: Correct Option (c)

The Indian Constitution was adopted on 26 Nov 1949 and came into force on 26 Jan 1950 (Republic Day).

Question 137: Correct Option (a)

Articles 14-18 grant equality before law, prohibition of discrimination, and abolition of untouchability.

Question 138: Correct Option (c)

The maximum strength of Lok Sabha permitted by the Constitution is 550 members.

Question 139: Correct Option (a)

PIL allows any individual or organization to file a petition in court for public good.

Question 140: Correct Option (b)

Article 17 abolishes Untouchability and forbids its practice in any form.

Question 141: Correct Option (b)

Governor-General Lord Cornwallis introduced the Permanent Settlement in Bengal, Bihar, and Odisha.

Question 142: Correct Option (b)

Charles Wood's Despatch laid down the blueprint for modern educational policy in India.

Question 143: Correct Option (c)

Solar, wind, tidal, and geothermal energy are non-conventional, renewable sources.

Question 144: Correct Option (b)

Population density of India rose to 382 persons per square kilometer in 2011.

Question 145: Correct Option (b)

Supreme Court ruled that Right to Life under Art. 21 includes right to enjoyment of pollution-free water and air.

Question 146: Correct Option (b)

Jyotirao Phule established Satyashodhak Samaj (Truth Seekers' Society) to fight caste discrimination.

Question 147: Correct Option (a)

Odisha is the largest producer of iron ore in India, contributing over half of national production.

Question 148: Correct Option (c)

The Speaker of Lok Sabha presides over a joint sitting summoned by the President under Art. 108.

Question 149: Correct Option (b)

Netaji Subhas Chandra Bose took over leadership of INA (Azad Hind Fauj) in Singapore in 1943.

Question 150: Correct Option (b)

The PoA Act was passed by Indian Parliament in 1989 to prevent atrocities against SC/ST members.